

Machine Learning Capstone — Review 1

Model Development & Evaluation Across Regression and Classification Tracks

PROJECT TEAM

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Objectives & Track Summary

REGRESSION TRACK

DATASET

RT_IoT2022 (IoT network traffic)

TASK

Predict flow_duration (seconds), a continuous target

CLEANED SIZE

117,922 rows × 53 leakage-free features

TRAIN / TEST SPLIT

94,337 / 23,585 rows (80/20, random_state=42)

MODELS EVALUATED

10: Linear, Ridge, Lasso, ElasticNet, Polynomial, Decision Tree, Random Forest, Gradient Boosting, SVR, KNN

CLASSIFICATION TRACK

DATASET

Dry Bean Dataset (UCI Machine Learning Repository)

TASK

Predict bean cultivar among 7 classes

CLEANED SIZE

13,543 rows (68 duplicates removed) × 17 features

TRAIN / TEST SPLIT

10,834 / 2,709 rows (80/20, stratified, random_state=42)

MODELS EVALUATED

5 (Part A): Logistic Regression, KNN, Decision Tree, Naive Bayes (Gaussian), SVM (linear & RBF)

EVALUATION PROTOCOL — COMMON TO BOTH TRACKS

- Deterministic 80/20 splits; all scalers and encoders are fit strictly on training data only.
- Hyperparameters tuned via GridSearchCV with cross-validation performed on training data only.
- Every model is evaluated exactly once on its held-out test set — no test-set model selection.
- Every metric, parameter and figure in this deck is transcribed directly from executed notebook output.

Dataset & Preprocessing

DATASET FACTS

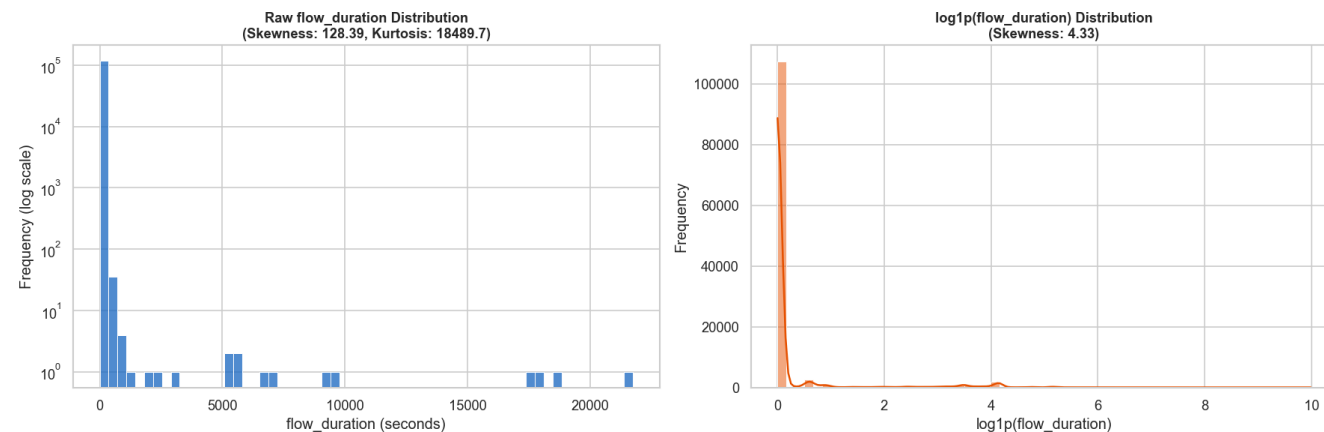
- RT_IoT2022 IoT network-traffic dataset
- Raw shape: 123,117 rows × 85 columns, zero missing values
- Target: flow_duration (seconds) — continuous, extremely right-skewed
- Skewness = 128.39, Kurtosis = 18,489.7 (full cleaned dataset)
- Median duration $\approx 4 \mu\text{s}$; maximum $\approx 21,728 \text{ s}$ (≈ 6 hours)

CLEANING, LEAKAGE PREVENTION & SPLIT

- Dropped 'Unnamed: 0' (row index) and 'bwd_URG_flag_count' (zero variance)
- Removed 5,195 duplicate rows (4.22%) $\rightarrow$ 117,922 rows × 83 columns
- Purged 30 leakage columns: 15 IAT timing, 5 active, 5 idle, 4 rate-derived, + Attack_type label $\rightarrow$ 53 leakage-free features
- Engineered total_packets = fwd_pkts_tot + bwd_pkts_tot ($r = 0.39$ with target)
- 80/20 split (seed=42): 94,337 train / 23,585 test rows (JSON-persisted indices)

OBSERVATION

flow_duration spans microseconds to hours in the same column. 5-fold CV on training data showed inverting a log1p transform back to raw seconds causes catastrophic variance blowout for linear models, so all 10 regression models are trained and evaluated on the raw target scale.



Actual output — regression_preprocessing.ipynb, raw vs. log1p target distribution

Dataset & Preprocessing

DATASET FACTS

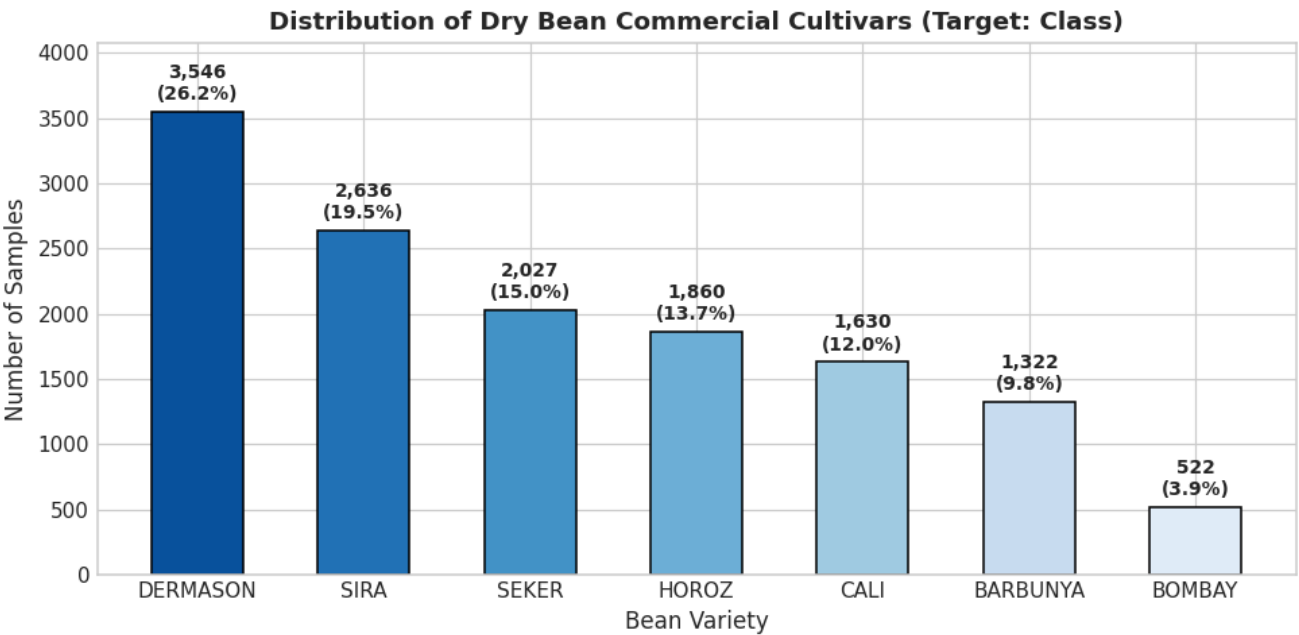
- Dry Bean Dataset (UCI Machine Learning Repository)
- Raw shape: 13,611 rows × 17 columns, zero missing values
- Target: Class — 7 cultivars (BARBUNYA, BOMBAY, CALI, DERMASON, HOROZ, SEKER, SIRA)
- 16 morphological features: size (Area, Perimeter, axis lengths) + shape (Compactness, Eccentricity, Shape Factors 1–4)
- Class imbalance: DERMASON largest (26.2%), BOMBAY smallest (3.9%)

CLEANING, LEAKAGE PREVENTION & SPLIT

- Removed 68 exact duplicates → 13,543 rows × 17 columns
- Engineered Area_Perimeter_Ratio = Area / Perimeter
- Target label-encoded to integers 0–6 via LabelEncoder
- Stratified 80/20 split (seed=42): 10,834 train / 2,709 test rows
- StandardScaler fit strictly on train, then applied to test

OBSERVATION

Because accuracy alone can hide poor minority-class performance, all 5 classifiers are additionally evaluated with weighted and macro F1-score, per-class precision/recall, and ROC-AUC — not accuracy alone.



Actual output — classification.ipynb, target class distribution

Linear Regression

HOW IT WORKS

Ordinary Least Squares fits a single linear hyperplane that minimizes the sum of squared residuals between predicted and actual flow duration. Categorical features (proto, service) are one-hot encoded and numeric features standardized before fitting. It assumes a linear relationship between packet/header volume and duration, with no thresholds or interactions.

ACTUAL CONFIGURATION (from notebook)

- LinearRegression() — default parameters
- Preprocessing: OneHotEncoder (cat) + StandardScaler (num)
- 53 leakage-free features (timing/rate columns purged)
- Trained on 94,337 rows, evaluated once on 23,585-row test set

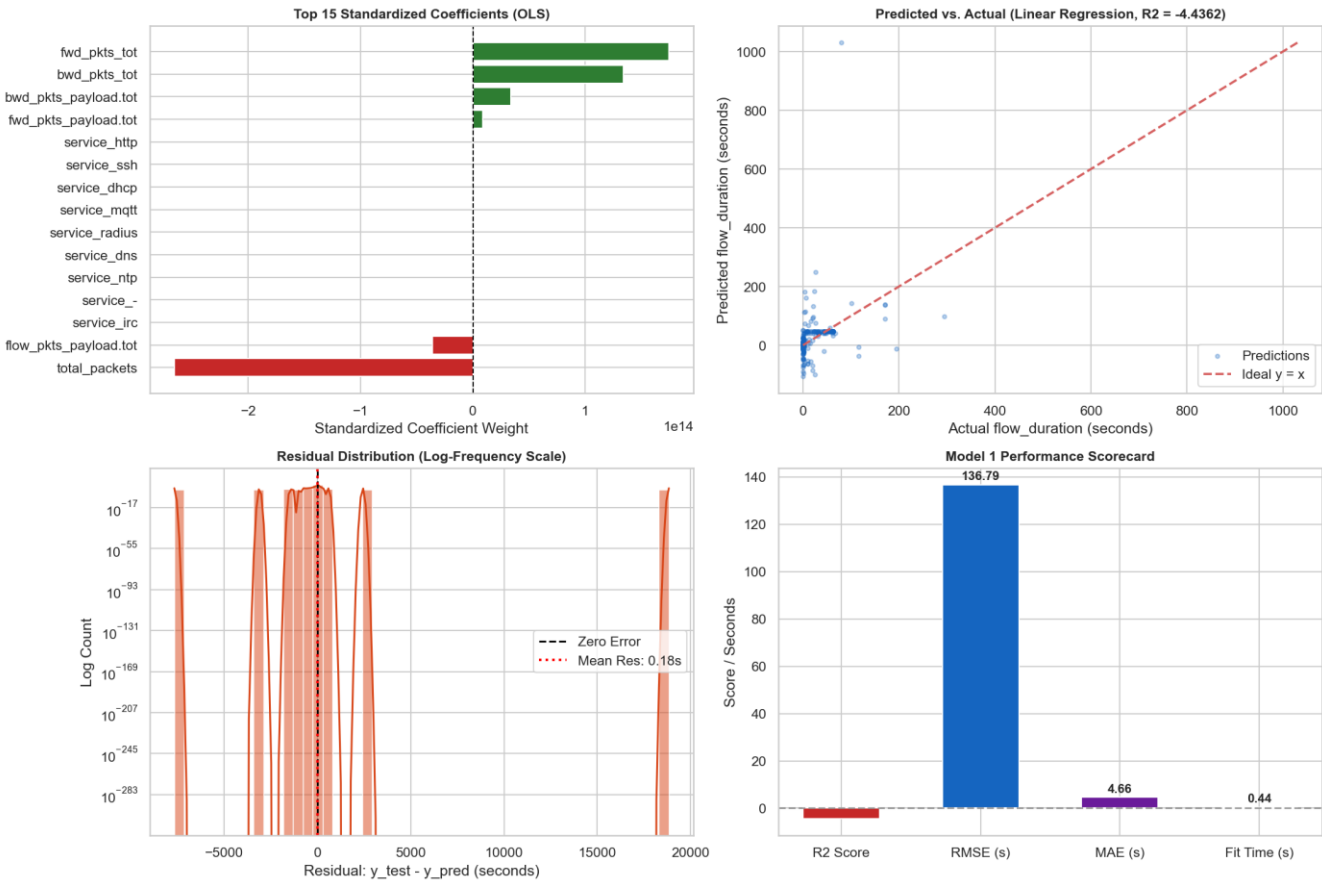
HELD-OUT TEST PERFORMANCE

Metric	R ²	RMSE (s)	MAE (s)
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Test (held-out)	-4.4362	136.7931	4.6552
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OBSERVATION

Linear Regression produced a strongly negative R² on the held-out test set, performing far worse than simply predicting the mean flow duration. This confirms the relationship between packet volume and duration is non-linear.



Actual output — regression.ipynb, Model 1 diagnostic suite (coefficients, predicted-vs-actual, residuals, scorecard)

Ridge Regression

HOW IT WORKS

Ridge regression adds an L2 penalty on coefficient magnitudes to the OLS objective, shrinking weights to reduce variance and stabilize collinear features such as header-size totals. It is still a linear model, so it can only rescale — not reshape — the OLS solution.

ACTUAL CONFIGURATION (from notebook)

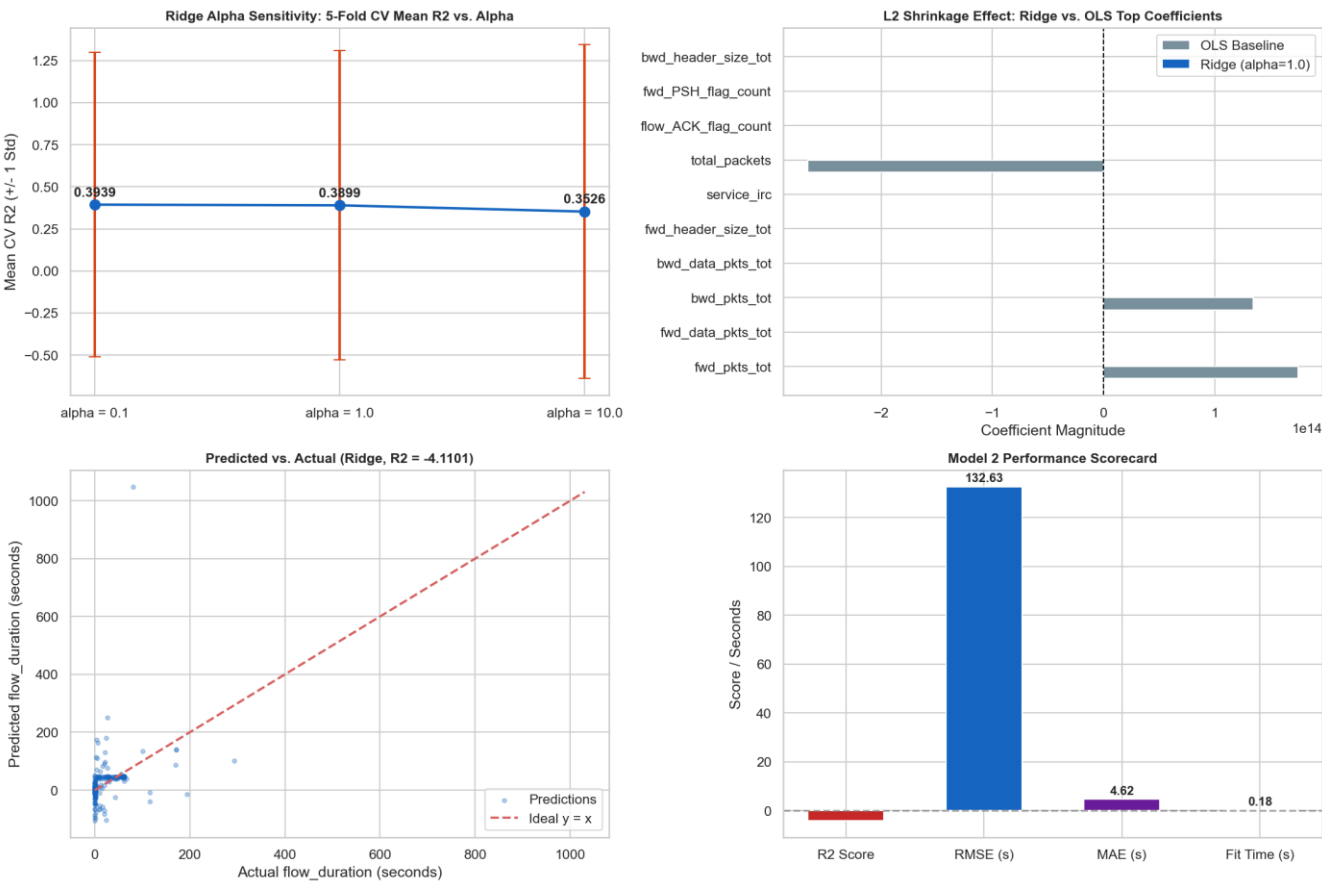
- Ridge(alpha=1.0, random_state=42) — official baseline
- Alpha sensitivity checked at 0.1 / 1.0 / 10.0 via 5-fold CV (train only)
- GridSearchCV(alpha=[0.1,1.0,10.0], cv=5) → best alpha=0.1 (CV R²=0.2501)
- Preprocessing: OneHotEncoder + StandardScaler

HELD-OUT TEST PERFORMANCE

Metric	R ²	RMSE (s)	MAE (s)
Baseline (α=1.0)	-4.1101	132.6261	4.6168
Tuned (α=0.1)	-4.3380	135.5515	4.6621

OBSERVATION

Ridge's L2 shrinkage cut RMSE by about 4 seconds versus OLS, but R² stayed strongly negative. The formally tuned alpha (selected by 5-fold CV) actually performed worse on the held-out test set — CV improvement did not carry over.



Lasso Regression

HOW IT WORKS

Lasso regression uses an L1 penalty that can shrink coefficients exactly to zero, performing automatic feature selection. At high regularization strength it aggressively discards features, which can also discard information the model actually needs.

ACTUAL CONFIGURATION (from notebook)

- Lasso(alpha=1.0, max_iter=2500, random_state=42) — official baseline
- Sparsity checked at alpha = 0.01 / 0.1 / 1.0 (training data only)
- At alpha=1.0: 46 of 64 encoded coefficients zeroed (18 active)
- Preprocessing: OneHotEncoder + StandardScaler

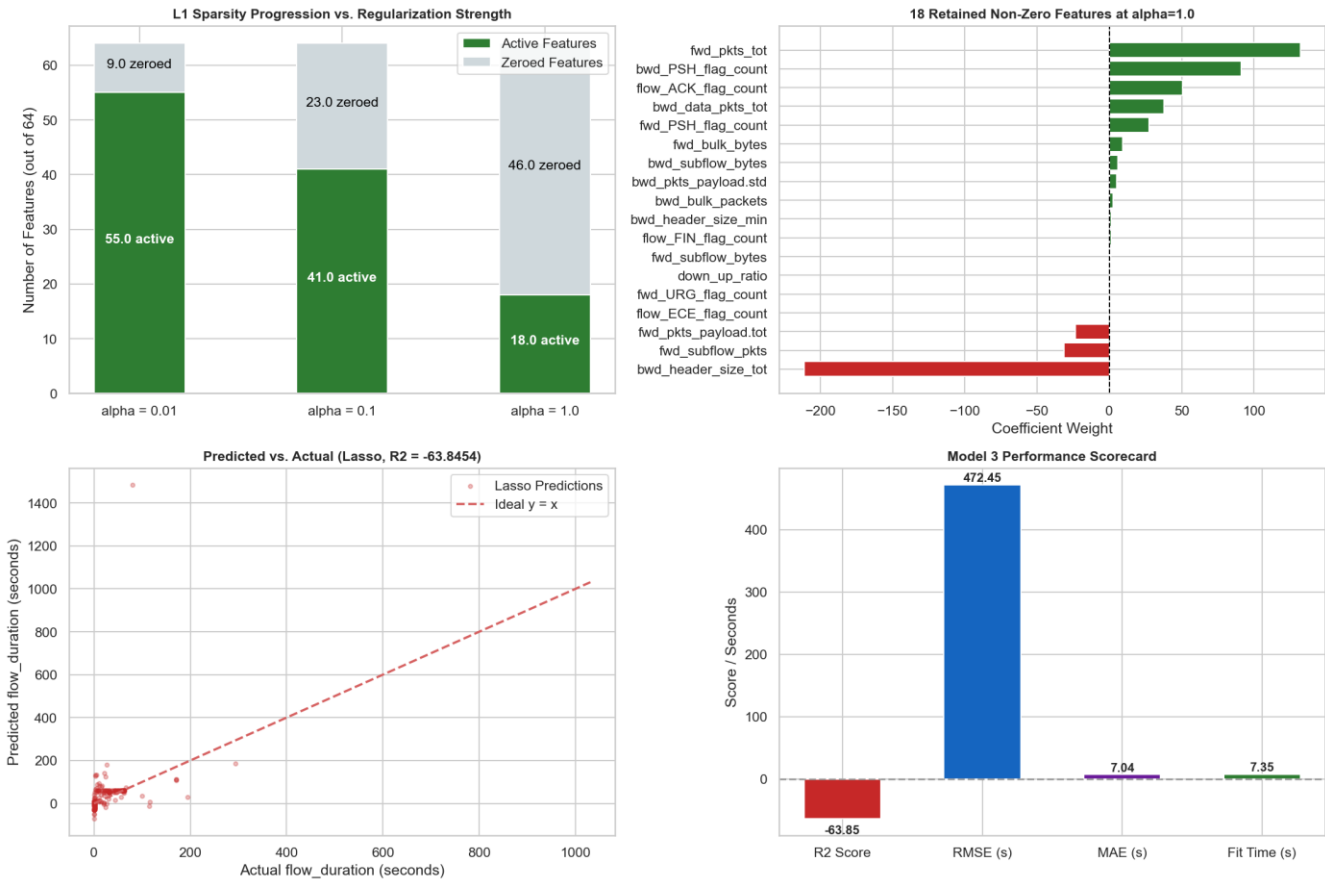
HELD-OUT TEST PERFORMANCE

Metric	R ²	RMSE (s)	MAE (s)
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Test (held-out)	-63.8454	472.4483	7.0382
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OBSERVATION

At alpha=1.0, Lasso zeroes out 72% of all features, discarding volumetric predictors the model needs. This produces the second-worst test R² of all ten regression baselines.



Actual output — regression.ipynb, Model 3 diagnostic suite (sparsity progression, predicted-vs-actual, scorecard)

ElasticNet Regression

HOW IT WORKS

ElasticNet linearly combines the L1 and L2 penalties, blending Lasso's feature selection with Ridge's coefficient shrinkage in a single tunable model.

ACTUAL CONFIGURATION (from notebook)

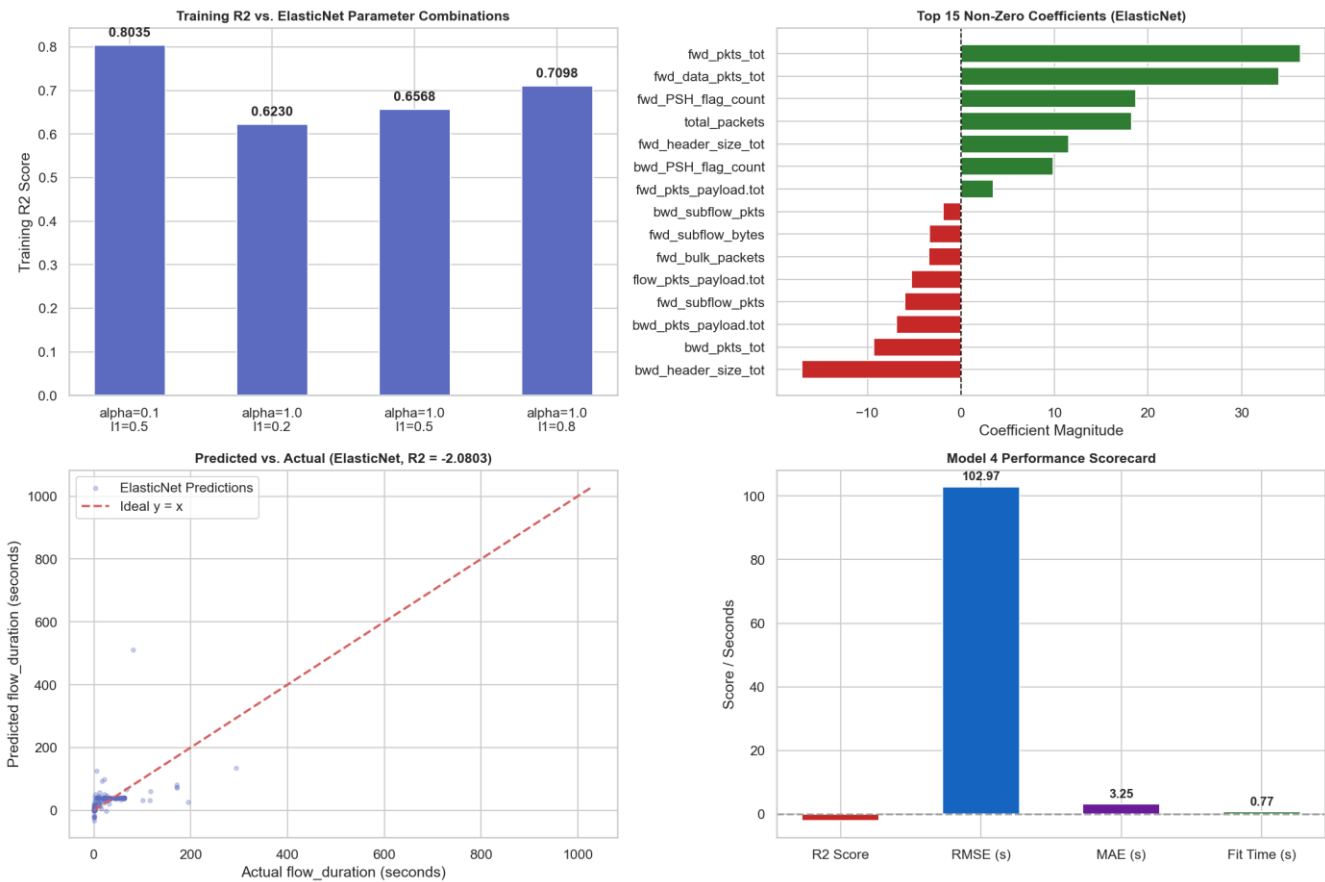
- ElasticNet(alpha=1.0, l1_ratio=0.5, max_iter=2500, random_state=42)
- Parameter behaviour explored at 3 additional (α , l1_ratio) pairs on training data
- Preprocessing: OneHotEncoder + StandardScaler

HELD-OUT TEST PERFORMANCE

Metric	R ²	RMSE (s)	MAE (s)
Test (held-out)	-2.0803	102.9699	3.2478

OBSERVATION

ElasticNet is the best-performing regularized linear model, cutting test MAE to 3.25 s and outperforming OLS, Ridge, Lasso and SVR — but its test R² remains negative, since it is still fundamentally linear.



Actual output — regression.ipynb, Model 4 diagnostic suite (parameter grid, predicted-vs-actual, scorecard)

Polynomial Regression

HOW IT WORKS

Polynomial regression expands a compact subset of four volumetric features into quadratic interaction terms, then fits ordinary least squares on the expanded set. It is restricted to 4 features to avoid a combinatorial blow-up of the full 53-feature space.

ACTUAL CONFIGURATION (from notebook)

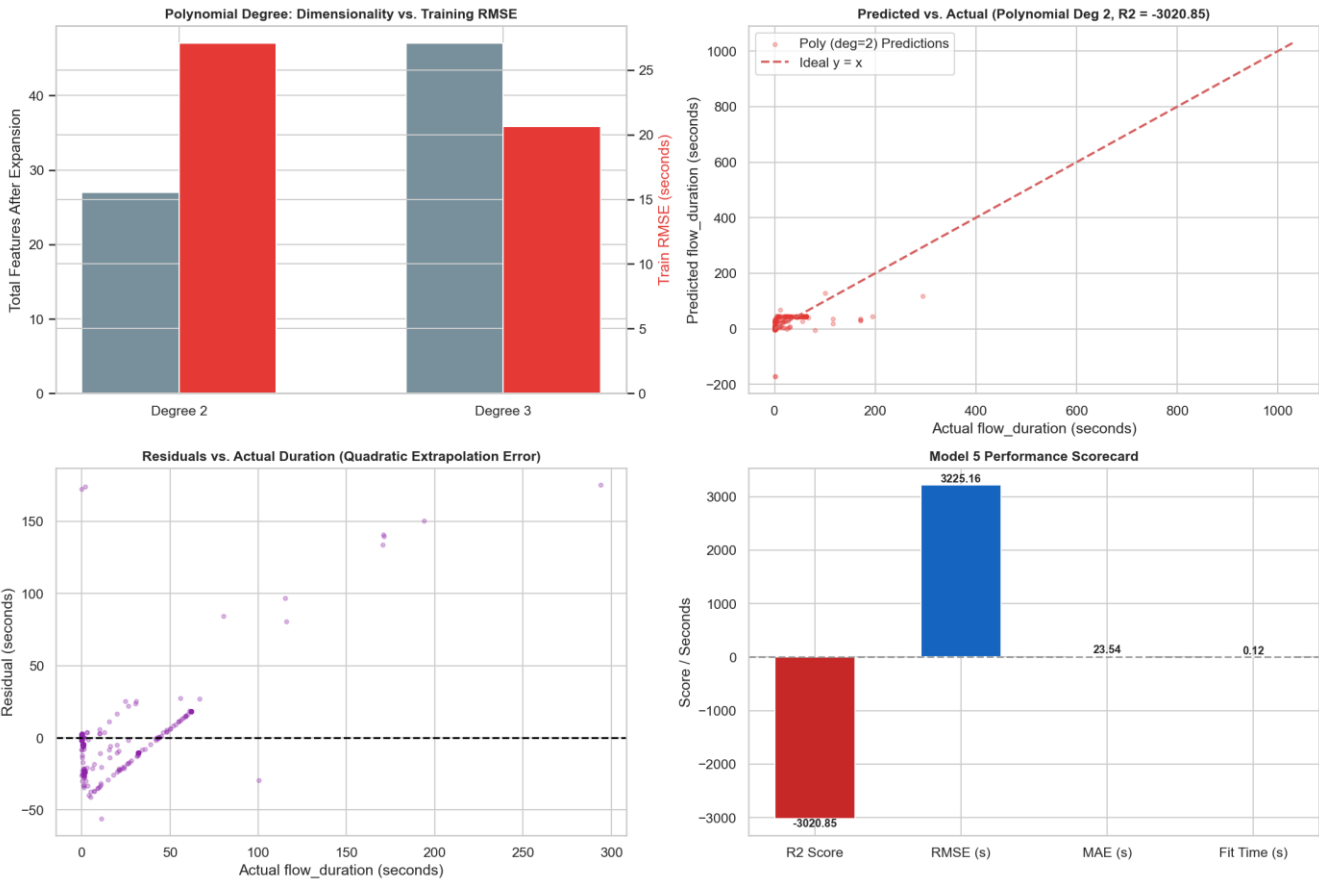
- Features expanded: fwd_pkts_tot, bwd_pkts_tot, fwd_header_size_tot, bwd_header_size_tot
- PolynomialFeatures(degree=2, include_bias=False) → 27 features
- LinearRegression() on expanded features — Degree 2 is the official baseline
- Degree 3 (47 features) explored only on training data for comparison

HELD-OUT TEST PERFORMANCE

Metric	R ²	RMSE (s)	MAE (s)
Train (deg=2)	0.9621	27.06	—
Test (held-out)	-3020.8537	3225.1623	23.5377

OBSERVATION

Despite explaining over 96% of variance on training data, unregularized quadratic terms extrapolate catastrophically once test samples fall slightly outside the training range — the worst result of all ten regression models.



Actual output — regression.ipynb, Model 5 diagnostic suite (degree comparison, predicted-vs-actual, scorecard)

Decision Tree Regressor

HOW IT WORKS

A decision tree recursively splits the feature space into regions, predicting the mean target value within each region. It naturally captures piecewise, non-linear relationships and step-function thresholds without feature scaling.

ACTUAL CONFIGURATION (from notebook)

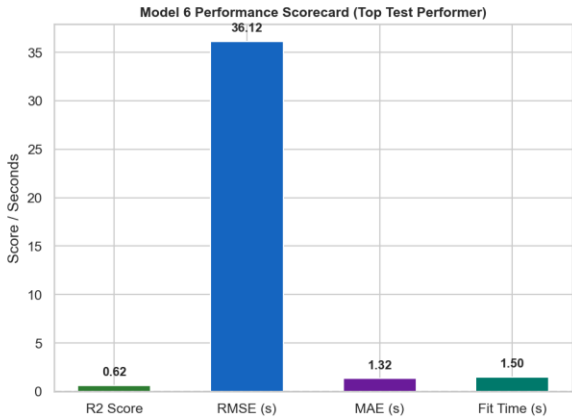
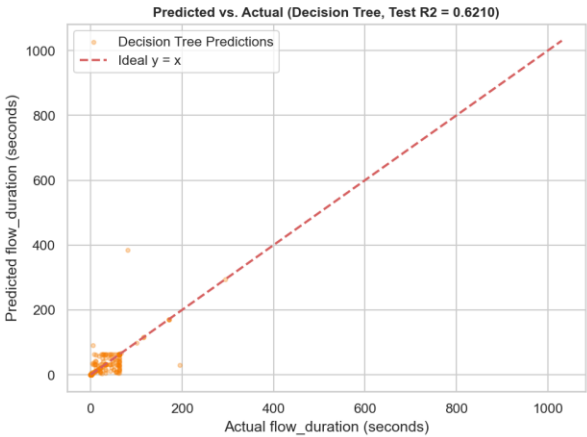
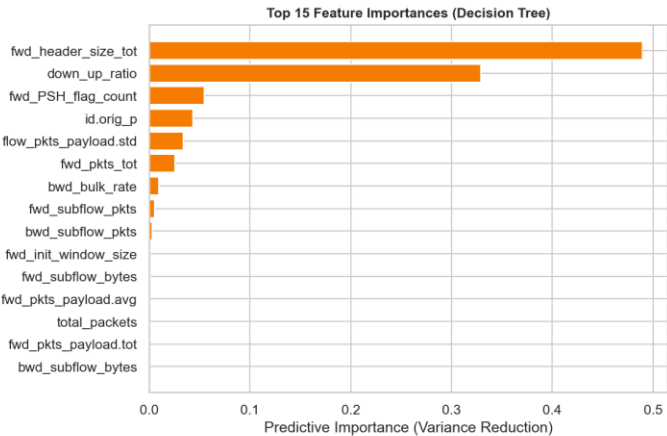
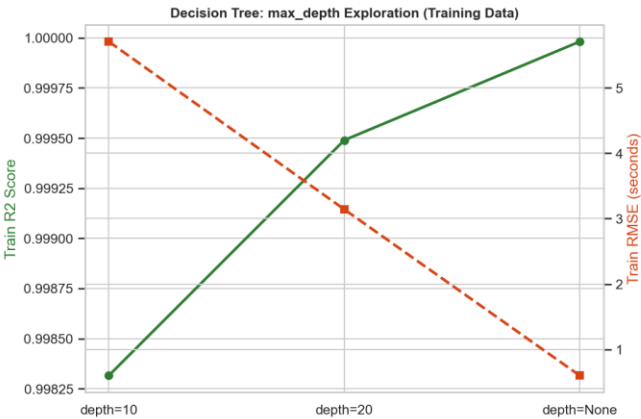
- DecisionTreeRegressor(random_state=42) — unconstrained depth (official baseline)
- max_depth ∈ {10, 20, None} explored only on training data
- Preprocessing: OneHotEncoder + passthrough numerics

HELD-OUT TEST PERFORMANCE

Metric	R ²	RMSE (s)	MAE (s)
Test (held-out)	0.6210	36.1167	1.3245
5-Fold Stratified CV	0.6821 ± 0.1795	63.31 ± 31.96	1.83 ± 0.57

OBSERVATION

Decision Tree is the top-performing model in the entire regression track — the only baseline with a strongly positive test R². Recursive partitioning matches the discrete handshake/protocol-state structure of network flows far better than any linear or margin-based model.



Actual output — regression.ipynb, Model 6 diagnostic suite (depth exploration, feature split behaviour, predicted-vs-actual, scorecard)

Random Forest Regressor

HOW IT WORKS

Random Forest averages predictions across many decision trees, each trained on a bootstrap sample with a randomized subset of features at each split. Averaging decorrelated trees reduces variance relative to a single tree.

ACTUAL CONFIGURATION (from notebook)

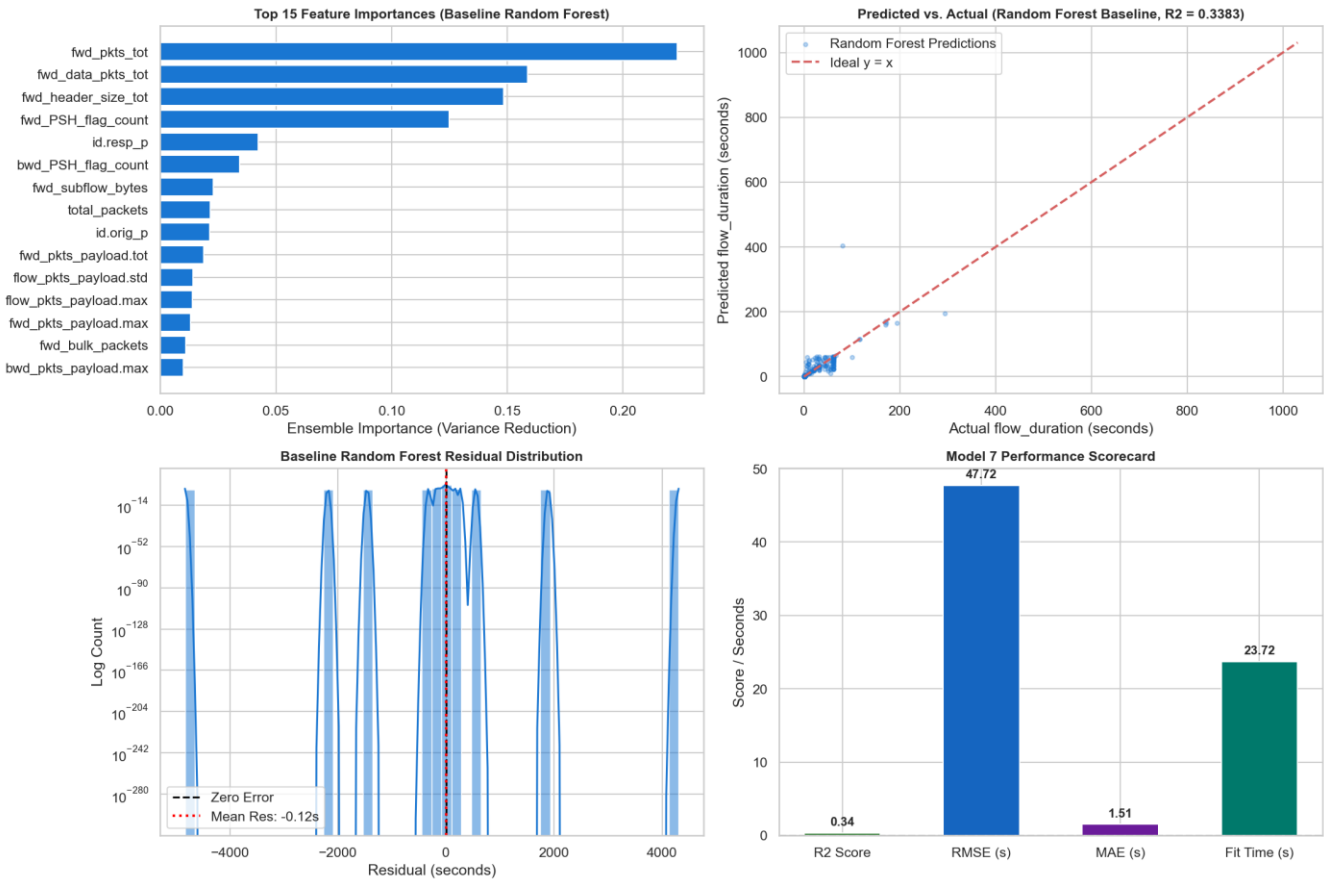
- Baseline: RandomForestRegressor(*n_estimators*=100, *random_state*=42)
- GridSearchCV(*cv*=5): *n_estimators* ∈ {50,100}, *max_depth* ∈ {15,25}
- Best params: *max_depth*=25, *n_estimators*=100 (best CV *R*² = 0.7512)

HELD-OUT TEST PERFORMANCE

Metric	R ²	RMSE (s)	MAE (s)
Baseline	0.3383	47.7241	1.5114
Tuned	0.3740	46.4201	1.4726
5-Fold CV (tuned)	0.6506 ± 0.2206	67.68 ± 32.67	1.90 ± 0.63

OBSERVATION

Tuning improved test *R*² from 0.3383 to 0.3740, making Random Forest the top ensemble model. Leaf averaging across 100 trees still smooths out rare, extreme-duration outliers, keeping it well behind the single Decision Tree.



Actual output — regression.ipynb, Model 7 diagnostic suite (feature importance, predicted-vs-actual, scorecard)

Gradient Boosting Regressor

HOW IT WORKS

Gradient Boosting builds trees sequentially: each shallow tree fits the residual errors left by the current ensemble, scaled by a learning rate, letting the model incrementally correct its own mistakes.

ACTUAL CONFIGURATION (from notebook)

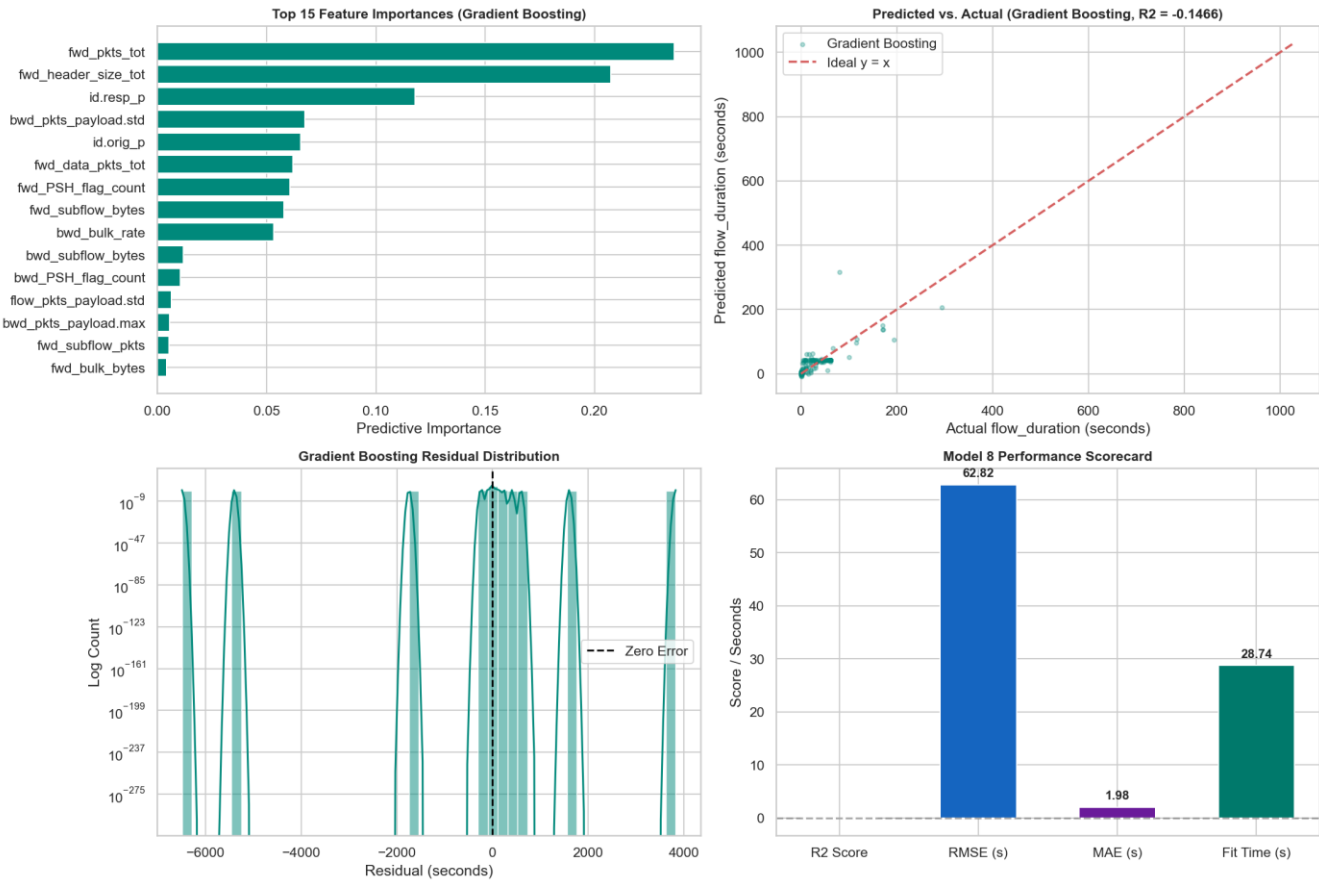
- Baseline: GradientBoostingRegressor(n_estimators=100, learning_rate=0.1, random_state=42)
- GridSearchCV(cv=5): learning_rate ∈ {0.05, 0.1}
- Best param: learning_rate=0.1 — identical to baseline (best CV R² = 0.6607)

HELD-OUT TEST PERFORMANCE

Metric	R ²	RMSE (s)	MAE (s)
Baseline / Tuned	-0.1466	62.8243	1.9760

OBSERVATION

Gradient Boosting reaches a strong cross-validation R² (0.6607) but a slightly negative test R² (-0.1466). Sequential fitting under squared-error loss lets early trees overcorrect on rare training outliers, hurting generalization to unseen tail events.



Actual output — regression.ipynb, Model 8 diagnostic suite (feature importance, predicted-vs-actual, scorecard)

Support Vector Regressor (SVR)

HOW IT WORKS

Support Vector Regression fits a linear function within an epsilon-insensitive margin, only penalizing predictions that fall outside that tube. A linear kernel was used on the complete, un-sampled training set, since non-linear kernels scale quadratically and were computationally infeasible at this size.

ACTUAL CONFIGURATION (from notebook)

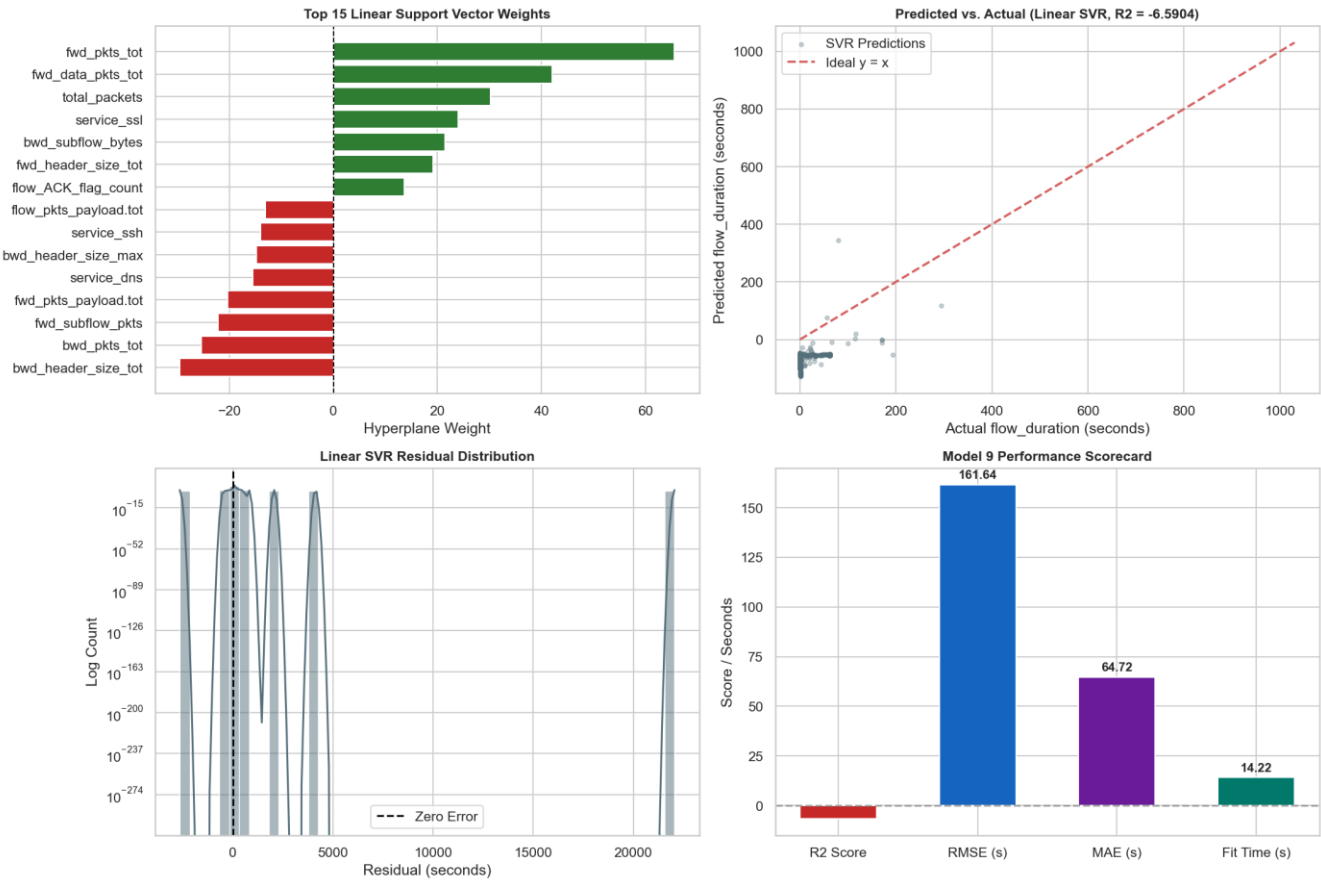
- SVR(kernel='linear', C=1.0, max_iter=2500)
- Trained on the full 94,337-row training set (no subsampling)
- Preprocessing: OneHotEncoder + StandardScaler

HELD-OUT TEST PERFORMANCE

Metric	R ²	RMSE (s)	MAE (s)
Test (held-out)	-6.5904	161.6391	64.7178

OBSERVATION

Linear SVR posts the second-worst MAE (64.72 s) of any regression model. A single linear margin cannot represent both microsecond handshakes and multi-hour sessions in the same tube.



Actual output — regression.ipynb, Model 9 diagnostic suite (support vector weights, predicted-vs-actual, scorecard)

K-Nearest Neighbors Regressor (KNN)

HOW IT WORKS

K-Nearest Neighbors predicts the average target value of the k closest training points in feature space, with no explicit training phase beyond storing the data.

ACTUAL CONFIGURATION (from notebook)

- `KNeighborsRegressor(n_neighbors=5)` — official baseline
- Preprocessing: `StandardScaler` (distance-based model)
- Sensitivity to $k \in \{3,5,10,20\}$ and scaling checked on a 20% training-only split

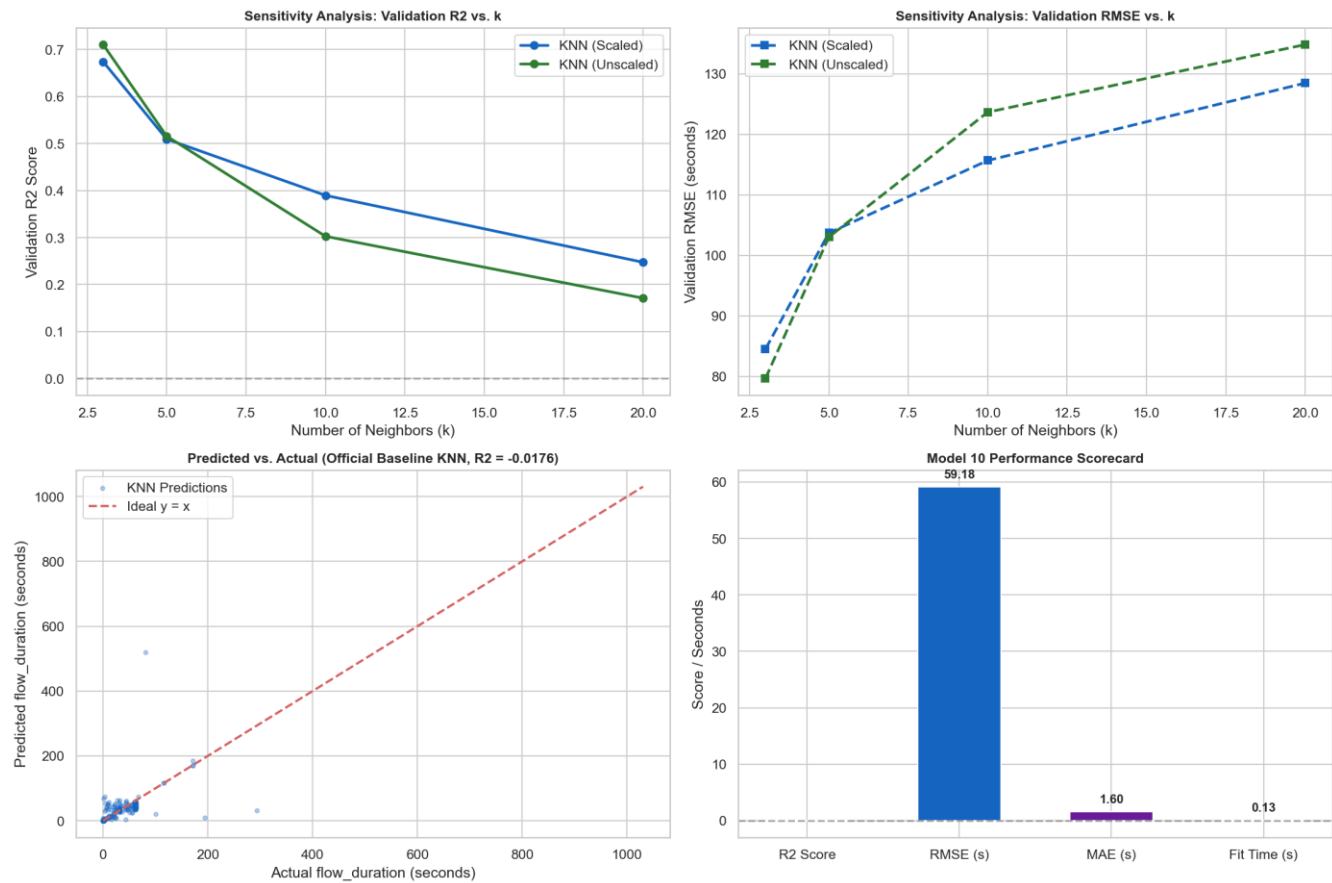
HELD-OUT TEST PERFORMANCE

Metric	R ²	RMSE (s)	MAE (s)
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Test (held-out)	-0.0176	59.1827	1.6035
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OBSERVATION

In 64-dimensional standardized space, distance concentration makes neighborhoods behave like a global average, pushing test R^2 near zero. Its MAE (1.60 s) stays competitive since most flows cluster around short durations.



Actual output — regression.ipynb, Model 10 diagnostic suite (k / scaling sensitivity, predicted-vs-actual, scorecard)

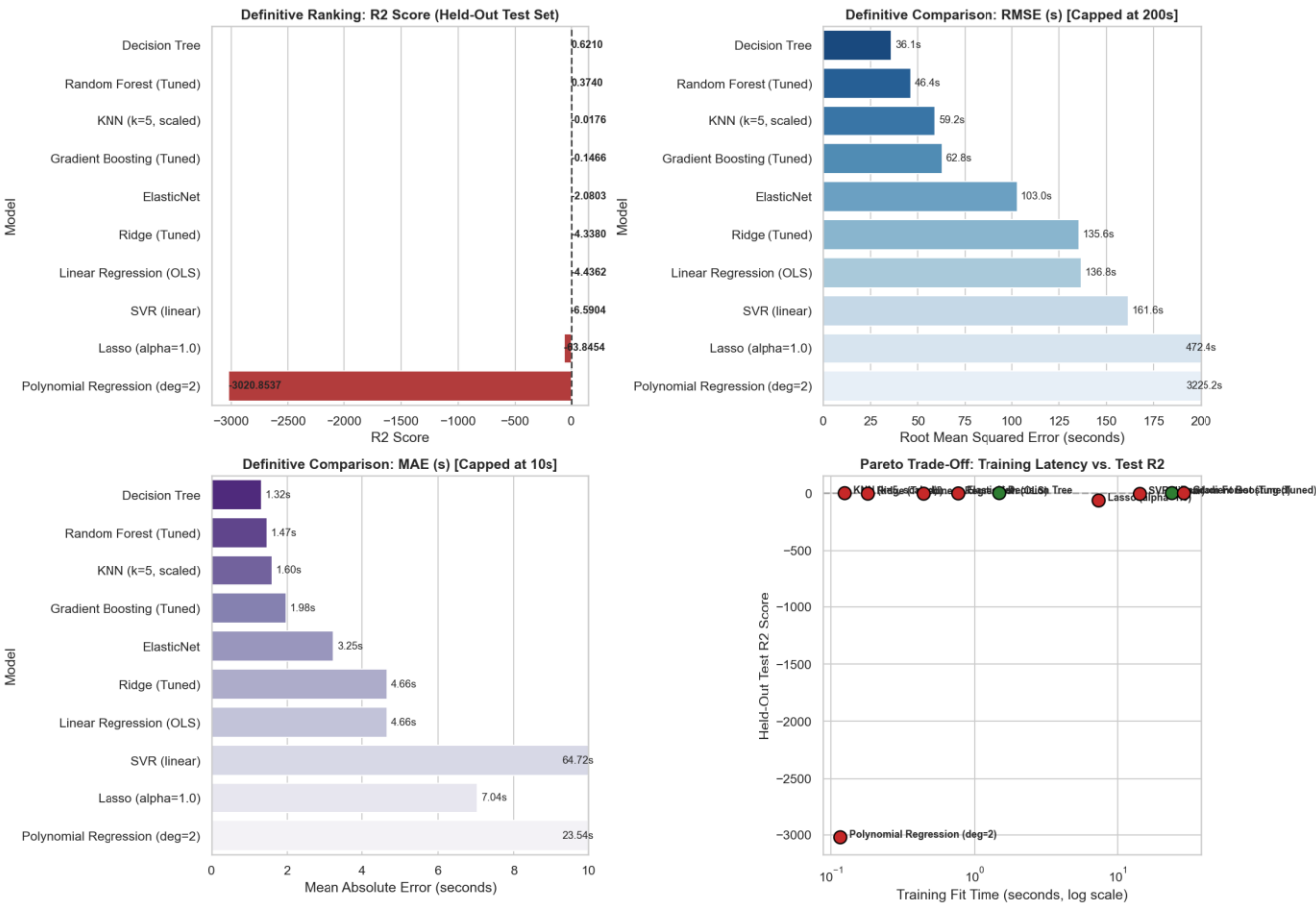
Final Model Comparison & Ranking

FINAL RANKING (HELD-OUT TEST SET)

Model	R ²	RMSE (s)	MAE (s)
Decision Tree	0.6210	36.12	1.32
Random Forest (Tuned)	0.3740	46.42	1.47
KNN (k=5, scaled)	-0.0176	59.18	1.60
Gradient Boosting (Tuned)	-0.1466	62.82	1.98
ElasticNet	-2.0803	102.97	3.25
Ridge (Tuned)	-4.3380	135.55	4.66
Linear Regression (OLS)	-4.4362	136.79	4.66
SVR (linear)	-6.5904	161.64	64.72
Lasso	-63.8454	472.45	7.04
Polynomial (deg=2)	-3020.85	3225.16	23.54

OBSERVATION

Decision Tree is the definitive top performer across the full regression benchmark — the only model with a strongly positive held-out R². Tuning meaningfully improved Random Forest (+0.0357 R²) but did not close the gap with the single tree. Formal cross-validated tuning did not help every model: tuned Ridge ($\alpha=0.1$) actually underperforms its untuned baseline ($\alpha=1.0$) on this held-out set, showing that CV improvement does not always generalize.



Logistic Regression

HOW IT WORKS

Multinomial Logistic Regression models the probability of each of the 7 bean cultivars using a softmax over linear combinations of standardized morphological features, producing linear decision boundaries between classes.

ACTUAL CONFIGURATION (from notebook)

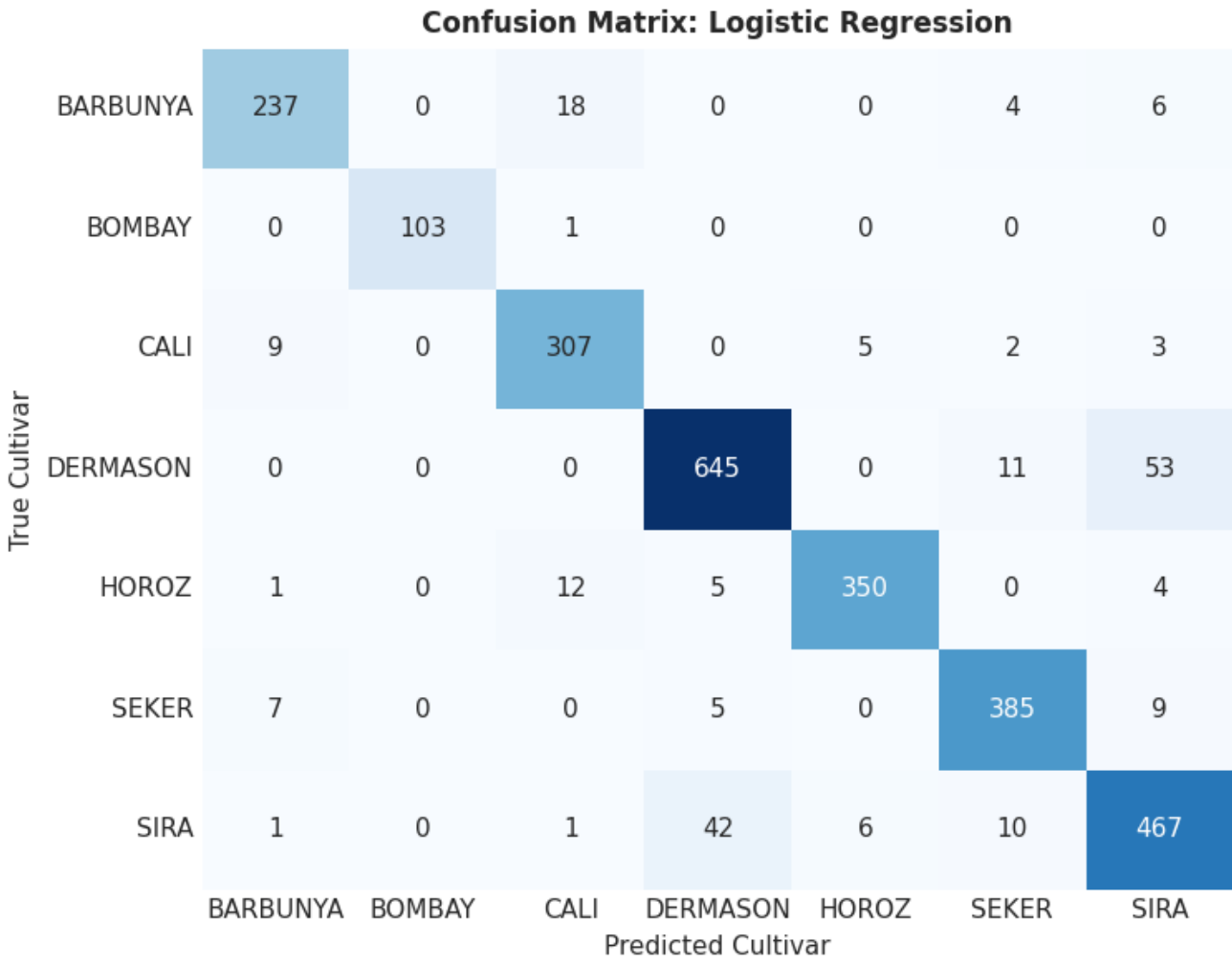
- LogisticRegression(max_iter=1000, random_state=42)
- GridSearchCV(cv=3, stratified): C ∈ {0.1, 1.0, 10.0}
- Best param: C=10.0 (best CV accuracy = 0.9244)
- Preprocessing: StandardScaler fit on train only

HELD-OUT TEST PERFORMANCE

Accuracy	Precision (w)	Recall (w)	F1 (w)	F1 (macro)	ROC-AUC
0.9206	0.9213	0.9206	0.9208	0.9311	0.9936

OBSERVATION

Logistic Regression is a strong linear baseline, second only to SVM in test accuracy. Its near-perfect separation of BOMBAY (100% precision) reflects that variety's distinctly larger size, while most confusion is between SIRA and DERMASON.



Actual output — classification.ipynb, held-out test confusion matrix

K-Nearest Neighbors (KNN)

HOW IT WORKS

KNN classifies each bean by majority vote among its k nearest neighbors in standardized feature space, based on Euclidean distance across all 17 features.

ACTUAL CONFIGURATION (from notebook)

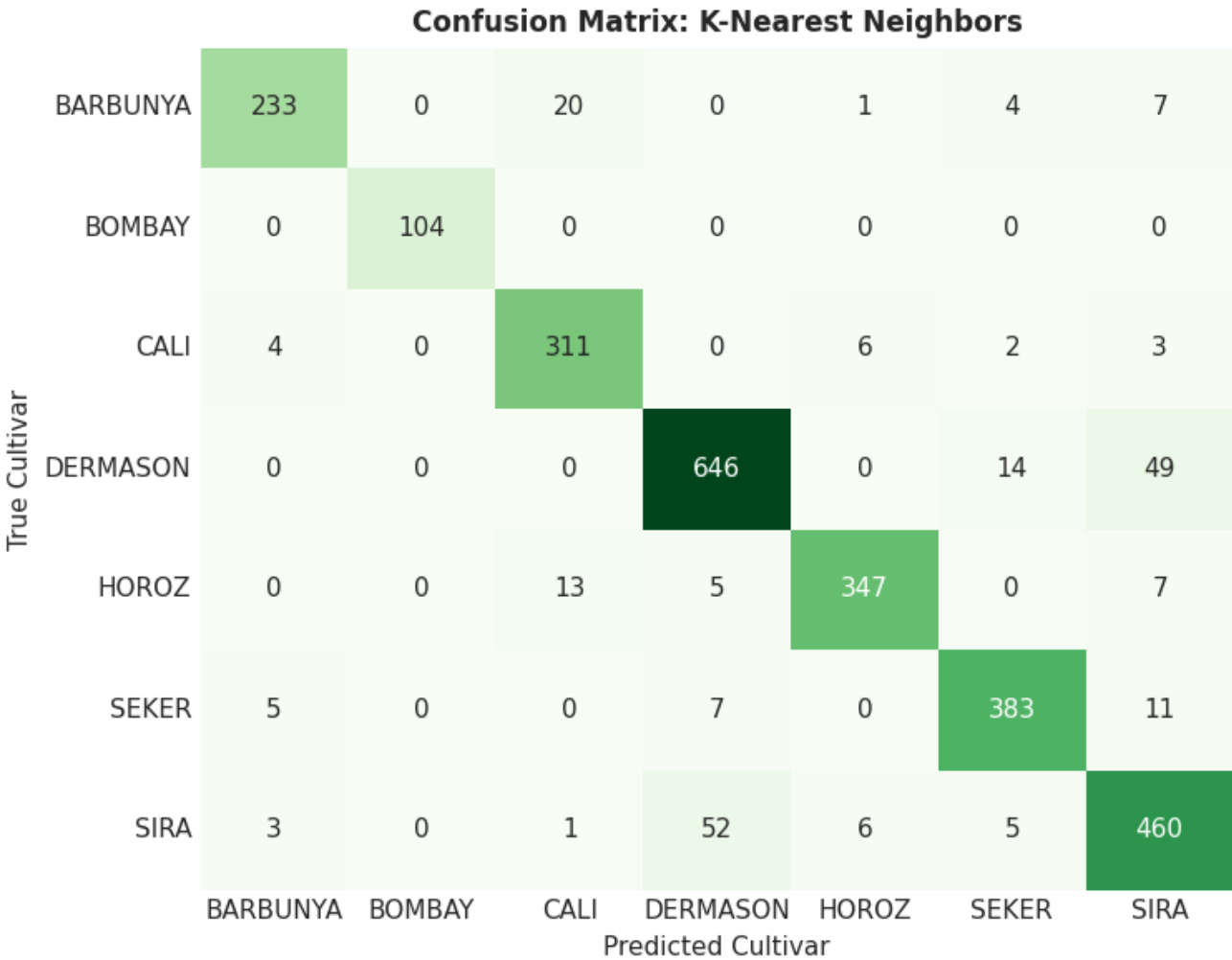
- KNeighborsClassifier()
- GridSearchCV(cv=3, stratified): n_neighbors ∈ {3,5,7,9,11,13}
- Best param: n_neighbors=13 (best CV accuracy = 0.9232)
- Preprocessing: StandardScaler fit on train only

HELD-OUT TEST PERFORMANCE

Accuracy	Precision (w)	Recall (w)	F1 (w)	F1 (macro)	ROC-AUC
0.9169	0.9177	0.9169	0.9171	0.9293	0.9880

OBSERVATION

k=13 balances bias and variance best under cross-validation. KNN resolves compact clusters like SEKER cleanly, with most remaining errors concentrated where SIRA and DERMASON densities overlap.



Actual output — classification.ipynb, held-out test confusion matrix

Decision Tree Classifier

HOW IT WORKS

A decision tree classifier recursively splits the feature space along single-feature thresholds, choosing splits that maximize Gini impurity reduction at each node, producing an interpretable set of if/then rules.

ACTUAL CONFIGURATION (from notebook)

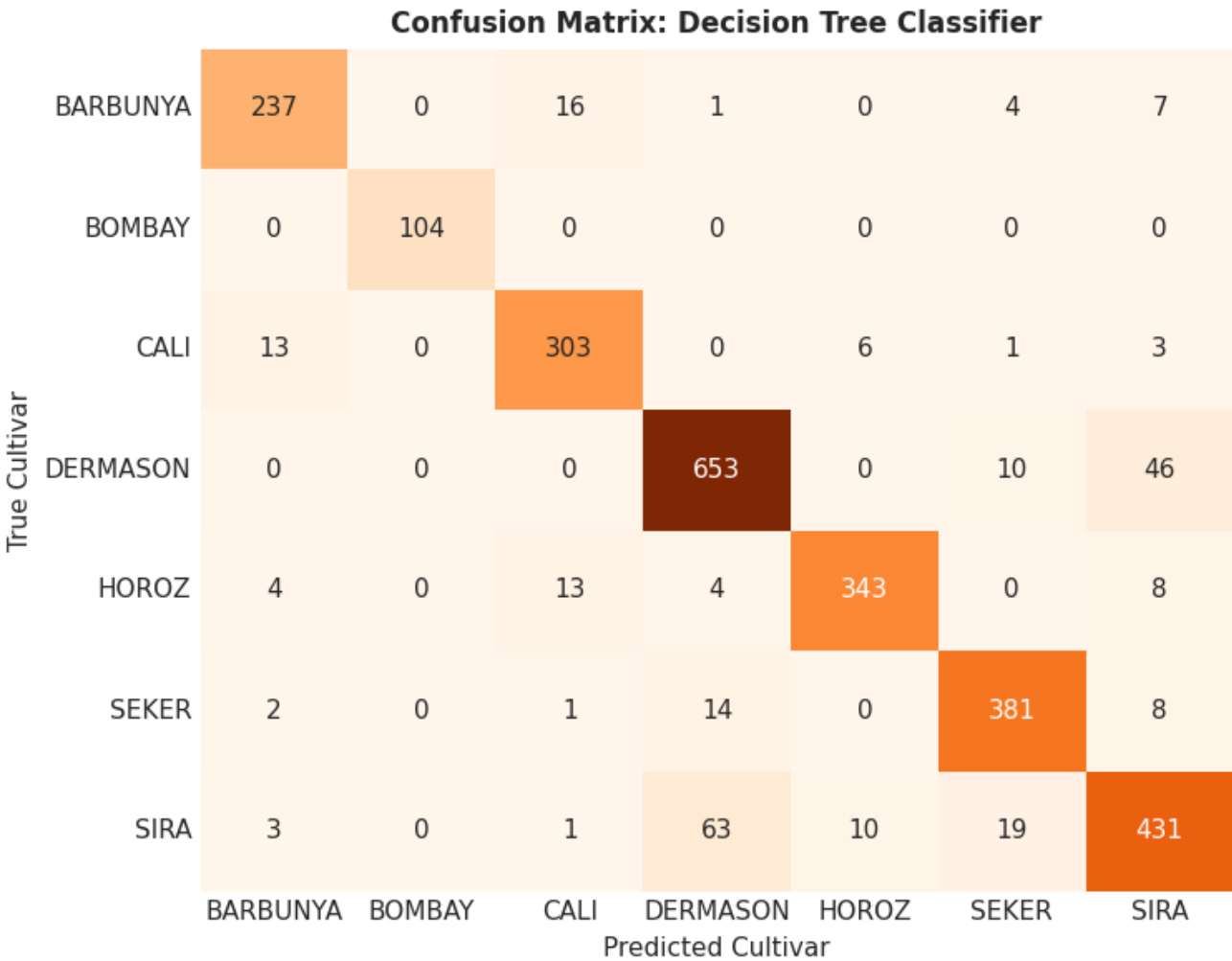
- DecisionTreeClassifier(random_state=42)
- GridSearchCV(cv=3, stratified): max_depth ∈ {6,8,10,12,None}, min_samples_split ∈ {2,5,10}
- Best params: max_depth=12, min_samples_split=10 (best CV accuracy = 0.9055)
- No feature scaling required (tree-based)

HELD-OUT TEST PERFORMANCE

Accuracy	Precision (w)	Recall (w)	F1 (w)	F1 (macro)	ROC-AUC
0.9051	0.9051	0.9051	0.9049	0.9187	0.9629

OBSERVATION

Decision Tree is the weakest of the five classifiers, though still above 90% accuracy. Its axis-aligned splits struggle with diagonal covariance between features like Area and Perimeter, but the tree remains fully interpretable.



Actual output — classification.ipynb, held-out test confusion matrix

Naive Bayes (Gaussian)

HOW IT WORKS

Gaussian Naive Bayes applies Bayes' theorem under the “naive” assumption that all features are conditionally independent given the class. It fits an independent 1-D Gaussian per feature per class, then picks the class with the highest posterior probability.

ACTUAL CONFIGURATION (from notebook)

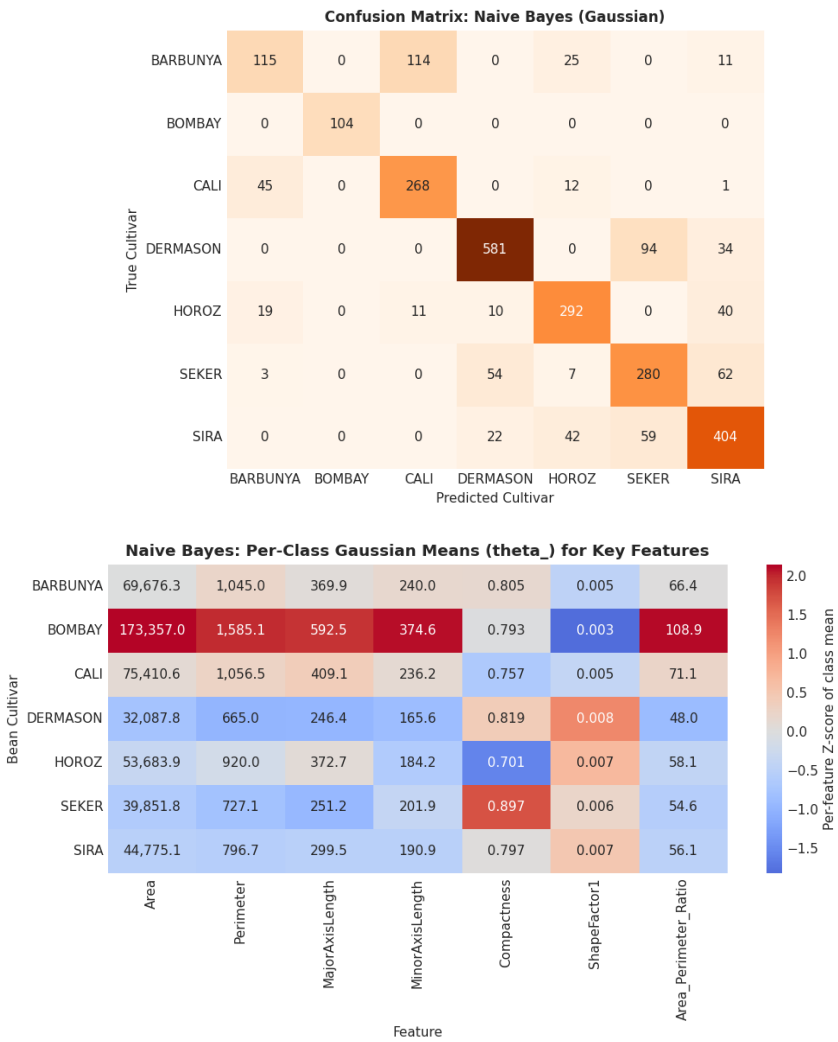
- GaussianNB() — default parameters (var_smoothing=1e-9)
- No hyperparameter search required per capstone guidelines for this model
- Trained on raw (unscaled) features — per-feature scaling does not change GaussianNB's decision boundary

HELD-OUT TEST PERFORMANCE

Accuracy	Precision (w)	Recall (w)	F1 (w)	F1 (macro)	ROC-AUC
0.7545	0.7556	0.7545	0.7521	0.7571	0.9622

OBSERVATION

Naive Bayes is the weakest of the five Part-A classifiers because its independence assumption is strongly violated — several size features correlate at $r \geq 0.95$ (Section 6). It still reaches a respectable ROC-AUC (0.9622), but double-counting correlated evidence drags accuracy down to 75.45%, with heavy confusion between BARBUNYA and CALI.



Actual output — classification.ipynb, confusion matrix (top) and per-class Gaussian means theta_ (bottom)

Support Vector Machine (SVM)

HOW IT WORKS

Support Vector Machine finds the hyperplane that maximizes the margin between classes; with the RBF kernel, it implicitly maps features into a higher-dimensional space where curved class boundaries become linearly separable.

ACTUAL CONFIGURATION (from notebook)

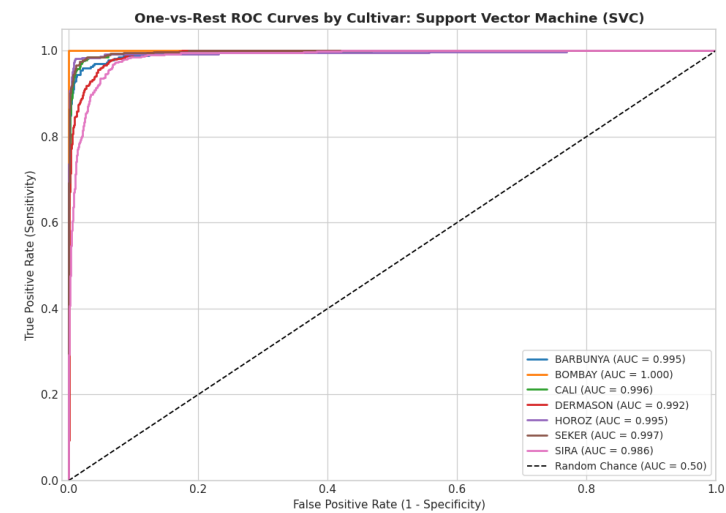
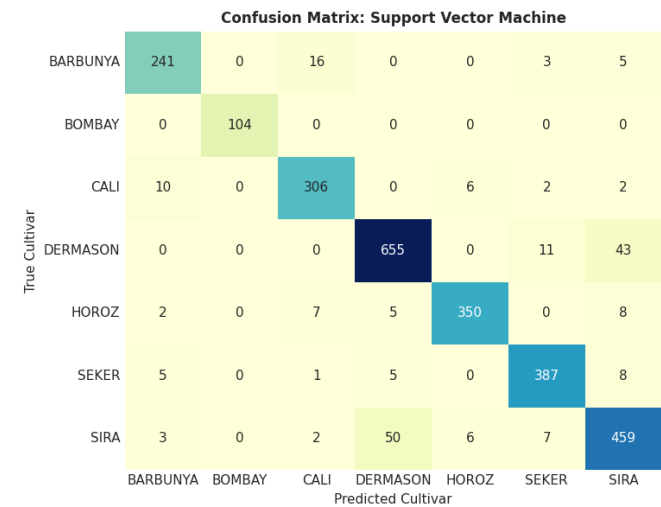
- SVC(probability=True, random_state=42)
- GridSearchCV(cv=3, stratified): C ∈ {1,10}, kernel ∈ {linear, rbf}
- Best params: C=10, kernel='rbf' (best CV accuracy = 0.9322)
- Preprocessing: StandardScaler fit on train only

HELD-OUT TEST PERFORMANCE

Accuracy	Precision (w)	Recall (w)	F1 (w)	F1 (macro)	ROC-AUC
0.9236	0.9237	0.9236	0.9236	0.9345	0.9931

OBSERVATION

SVM with an RBF kernel is the best-performing model in the classification track on every metric. The non-linear kernel resolves curved overlap between SIRA and DERMASON better than any linear model.



Actual output — classification.ipynb, confusion matrix (top) and one-vs-rest ROC curves (bottom)

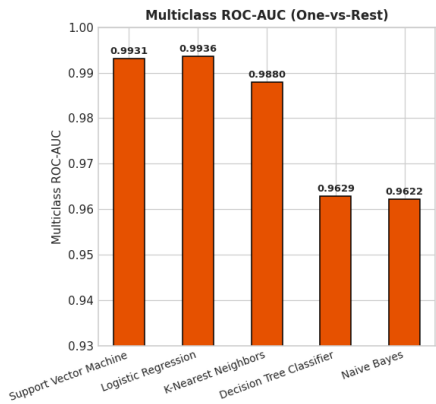
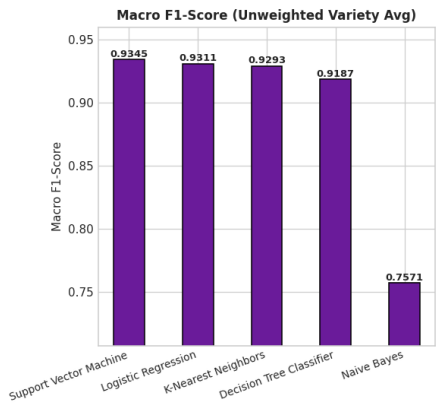
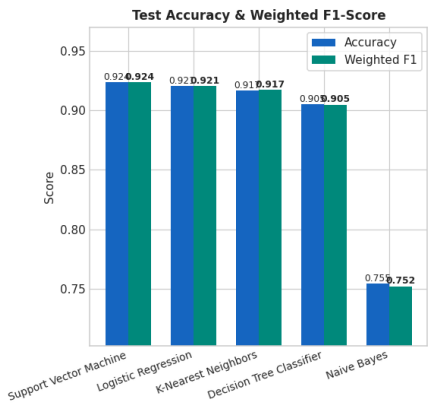
Final Model Comparison & Ranking

FINAL RANKING (HELD-OUT TEST SET)

Model	Accuracy	F1 (w)	F1 (macro)	ROC-AUC
SVM (RBF, C=10)	0.9236	0.9236	0.9345	0.9931
Logistic Regression	0.9206	0.9208	0.9311	0.9936
KNN (k=13)	0.9169	0.9171	0.9293	0.9880
Decision Tree	0.9051	0.9049	0.9187	0.9629
Naive Bayes (Gaussian)	0.7545	0.7521	0.7571	0.9622

OBSERVATION

Four of the five Part-A classifiers exceed 90% test accuracy, confirming that morphological features are highly discriminative for cultivar identification. SVM with an RBF kernel is the best overall performer on accuracy, weighted F1 and macro F1, narrowly ahead of Logistic Regression's strong linear baseline. Naive Bayes trails substantially (75.45%) because its conditional-independence assumption is violated by highly correlated size features — though it still posts a strong 0.9622 ROC-AUC, showing its class ranking stays reasonable even as its calibrated accuracy suffers.



Key Takeaways

REGRESSION TRACK

BEST MODEL

Decision Tree Regressor — $R^2 = 0.6210$, RMSE = 36.12 s, MAE = 1.32 s (the only baseline with a strongly positive test R^2).

WHY IT WINS

flow_duration is extremely right-skewed (skew = 128.39) and piecewise in nature; recursive partitioning matches this far better than any linear, regularized, or margin-based model.

ENSEMBLES

Tuned Random Forest is the best ensemble ($R^2 = 0.3740$) but leaf-averaging smooths rare, extreme-duration outliers, keeping it behind a single tree.

LINEAR MODELS

OLS, Ridge, Lasso, ElasticNet and linear SVR all post negative R^2 — the packet-volume-to-duration relationship is fundamentally non-linear.

CLASSIFICATION TRACK

BEST MODEL

Support Vector Machine (RBF, C=10) — Accuracy = 92.36%, Macro F1 = 0.9345, ROC-AUC = 0.9931 — best on every reported metric.

WHY IT WINS

The RBF kernel resolves curved, overlapping boundaries between visually similar cultivars (SIRA vs. DERMASON) better than a linear decision surface.

OVERALL TIER

4 of 5 Part-A classifiers exceed 90% test accuracy — morphological features (area, perimeter, shape factors) are highly discriminative for this task.

WEAKEST MODEL

Naive Bayes (Gaussian) trails at 75.45% accuracy — its independence assumption is violated by size features correlated at $r \geq 0.95$, though it still reaches a strong 0.9622 ROC-AUC.

CLASS IMBALANCE

BOMBAY (3.9% of data) is classified near-perfectly by every model due to its distinctly larger size; confusion concentrates in SIRA vs. DERMASON.

Both tracks followed a leakage-free, single held-out-evaluation protocol with cross-validated hyperparameter tuning — results here are reproducible end to end.